

Annotated Bibliography

Principal Grey-Boxing topics related to IBR/Power System

Zhang, T., Hao, Z., Ma, H., Yang, S., and Li, C. Dominant mode identification for grey-box grid-tied converters. *Electric Power Systems Research* 228 (3 2024), 110038

The DMI (Dominant Mode Identification) method aims to identify the system's dominant modes (critical eigenvalues) to assess its stability. Unlike approaches that require a complete predefined model structure, this method focuses on identifying these critical eigenvalues without needing detailed internal converter structure knowledge. It assumes the converter is a "black-box" while the grid is a "white-box" with a known Thévenin equivalent model.

The grey-box procedure is based on Frequency Sweeping (FS), injecting small perturbation voltage signals (Δv_1 and Δv_2 for dq-axis components, with positive and negative frequencies) and measuring current responses to obtain the black-box converter's admittance or impedance matrix. The closed-loop transfer function of the converter-grid system is then calculated using the grid's impedance and the converter's admittance.

To identify critical eigenvalues from this transfer function, the paper proposes using the Particle Swarm Optimization (PSO) algorithm. PSO is chosen for its robustness to measurement noise and for not requiring the assumed model order to be no less than the actual order (a known limitation of Vector Fitting (VF), which VF itself tries to overcome, and is thus explicitly avoided for this phase). The PSO algorithm optimizes the critical eigenvalues by minimizing a fitness function representing the error between measured transfer function values and those predicted by the PSO model, defining search ranges for the real and imaginary parts of the eigenvalues and associated residues.

Zong, H., Lyu, J., Wang, X., Zhang, C., Zhang, R., and Cai, X. Grey box aggregation modeling of wind farm for wideband oscillations analysis. *Applied Energy* 283 (2 2021), 116035

This work introduces a grey-box aggregation model for wind farms, focusing on wideband oscillation analysis and frequency-domain consistency. The methodology in-

volves modeling the equivalent impedance of a complete DFIG-based WTG (including main circuit, RSC, GSC controls, and PLL) as a single-machine grey-box model, while ignoring the mechanical part for electrical oscillation analysis. A crucial part of the method is parameter order reduction, as the initial grey-box impedance model contains about 20 unknown parameters (8 electrical, 12 control), which would make the identification problem "ill-conditioned". To optimize this process, a two-step quantitative evaluation procedure is used:

1. Trajectory sensitivity analysis (in the time domain): Evaluates the influence of each parameter on the system's dynamic response (e.g., active/reactive power), identifying "key" parameters and omitting those with negligible sensitivity.
2. Wideband characteristics analysis (in the frequency domain): Identifies specific frequency bands where controls have a greater impact. For example, RSC controls affect low-medium frequencies, while GSC control delay and the filter capacitor affect high frequencies. Based on this analysis, two reduced-order grey-box models are established: one for the low-medium frequency band (<200 Hz), considering 7 key control parameters, and one for the high frequency band (>200 Hz), considering 3 key control parameters and the filter capacitance (C_{filter}). The approach relies on Vector Fitting (VF) for parameter identification. VF is applied in two steps:
 - approximating measured frequency data into a standard rational function, which may involve a "trade-off" by ignoring weaker off-diagonal elements of some matrices to avoid excessive rational function order.
 - Optimizing unknown parameters through a two-step iterative process using nonlinear optimization tools. The need to pre-process raw frequency data by subtracting the equivalent impedance of passive components (transformers, collection network) is also mentioned.

Zhu, Y., Gu, Y., Li, Y., and Green, T. Participation Analysis in Impedance Models: The Grey-Box Approach for Power System Stability. *IEEE Transactions on Power Systems* 37, 1 (1 2022), 343–353

This article introduces an innovative grey-box approach for small-signal stability analysis of complex power systems. The term "grey-box" here signifies the ability to "look

inside" black-box models without full disclosure of internal details, facilitated by a "chain-rule" linking state, impedance, and parameter participation factors. The core of the approach is the impedance participation factor, defined as the negative of the conjugate transpose of the residue matrix of the whole-system admittance calculated at a specific eigenvalue. This definition establishes a direct link between an eigenvalue's sensitivity and an apparatus's impedance, allowing quantification of specific components' influence on overall stability. The grey-box approach is structured into three layers of increasing transparency:

1. Layer-1 (Minimum Transparency): Requires only apparatus impedance (Z_k) and the network's nodal admittance matrix (Y_{net}). It identifies potential participants in a given oscillation mode, using $p\lambda, Z_k Z_k(\lambda)$ as the primary index.
2. Layer-2 (Intermediate Transparency): Adds the assumption that apparatus impedance perturbations (ΔZ_k) are aligned with the original impedance ($\Delta Z_k \in Z_k$). It uses $\in \prec p\lambda, Z_k, Z_k(\lambda) \succ$ to provide information on a node's impact on system damping.
3. Layer-3 (Maximum Transparency): Leverages the sensitivity of an impedance to its internal parameters ($\partial Z_k / \partial \rho$) to propagate a parameter perturbation to an impedance perturbation and, finally, to an eigenvalue perturbation via the "chain-rule". This layer enables identification of the root-cause of instability and suggests which internal parameters should be re-tuned for system stabilization. Practical implementation involves obtaining data ($Z_k, \partial Z_k / \partial \rho$) via model-based routes (analytical models, linearization to obtain transfer functions) or measurement-based routes (disturbance injection, FFT for frequency spectra). Data are presented as frequency spectra to avoid symbolic calculations and ensure scalability.

Cecati, F., and Liserre, M. Interoperability Specifications for Multi-Vendor Converter-Dominated Grid: a Robust Stability Perspective. *IEEE Transactions on Smart Grid* (2025)

This work proposes a methodology for formulating interoperability specifications in multi-vendor converter-dominated grids, treating the problem as one of robust stability. The grey-box approach is nonlinear and aims to overcome the limitations of

conventional small-signal black-box models, which are accurate only around a nominal operating point and inaccurate across a wide range of operating conditions. The grey-box modeling strategy is based on integrating known nonlinear blocks (e.g., DC-side power conservation law, rotation matrices $T(\delta)$ and $T^{-1}(\delta)$) with linear blocks with unknown dynamics, representing control loops ($G_{ci}(s)$, $G_{dvc}(s)$, $G_{syn}(s)$, $G_{avc}(s)$). These unknown control blocks are modeled using frequency-dependent multiplicative uncertainties in the form $G(s) = \bar{G}(s)(1 + w\delta(s))$. Here, $\bar{G}(s)$ is a nominal transfer function (an initial estimate, often a PI controller with tuned gains), $\delta(s)$ is a frequency-dependent deviation (e.g., a high-pass filter to model greater uncertainty at high frequencies), and w quantifies the amount of uncertainty. The definition of $\bar{G}(s)$ and $\delta(s)$ is done loop-by-loop, adapting to each control loop's characteristics. The methodology for determining interoperability specifications involves three steps:

1. Stability analysis of the nominal grid model: Eigenvalue analysis is performed to ensure the nominal system (without uncertainty) is stable and well-damped.
2. Sensitivity analysis of the robust stability margin: Identifies which control loops are most critical for system robust stability in the presence of defined uncertainties.
3. Destabilizing perturbation analysis and uncertainty redesign: For critical control loops, the "destabilizing perturbation" (the smallest uncertainty value causing instability) is identified. The uncertainty is then redesigned (e.g., by reducing PLL tuning limits) until robust stability is guaranteed, providing stricter tuning guidelines for suppliers. These redefined limits constitute the "interoperability specifications".

Pan, Z., Tang, F., Zhang, Q., and Liu, Z. Grey Box Modeling Method of Grid-Connected Inverters Based on Parameter Sensitivity. 2025 8th International Conference on Energy, Electrical and Power Engineering, CEEPE 2025 (2025), 891–896

This article proposes a grey-box modeling method for grid-connected inverters, specifically designed to address commercial confidentiality and improve identification accuracy of unknown control parameters. Grey-box here refers to systems with known main circuit structures but unknown control parameters (such as k_{ip} , k_{ii} for the current loop and k_{ppll} , k_{ipll} for the PLL). The procedure begins with the analysis of the

mechanistic inverter model and small-signal AC-side impedance modeling. Main circuit and control equations, linearized around a steady-state operating point, are used to derive the AC-side impedance ($Z_{pcc,dq}$). Model validation is performed by comparing simulated impedance sweeps with theoretical results. A distinctive feature of this method is the detailed control parameter sensitivity analysis. Bode diagrams of AC-side impedance components (Z_{dd} , Z_{dq} , Z_{qd} , Z_{qq}) are observed as control parameters (k_{ip} , k_{ii}) vary. Frequencies where amplitude or phase show significant changes are identified as high-sensitivity frequencies. This analysis is crucial for optimal identification frequency and impedance element selection for each parameter, a key step to maximize identification accuracy (e.g., 80Hz for k_{ip} , 30Hz for k_{ii}). For parameter identification, the Particle Swarm Optimization (PSO) algorithm is employed. It is chosen for its rapid convergence, ease of tuning, and robust global search capabilities, suitable for nonlinear and multi-variable problems. The PSO fitness function is defined as the error between the theoretical impedance model (Z_{model}) and the impedance obtained via frequency sweeping from the actual model (Z_{target}). The process is iterative: if fitting accuracy is insufficient, identification frequencies and impedance elements are re-selected.

Li, Y., Wu, B., Li, J., Yue, W., and Wang, Y. A Grey Box Method for the Stability Analysis of Multi-Inverter-Fed Power Systems using Bode Diagrams

This study proposes a grey-box method for multi-inverter system stability analysis using Bode diagrams, aiming to overcome the limitations of the Impedance-Frequency (IF) method. These limitations include the IF method's implicit assumption of zero RHP poles in aggregated impedance (potentially incorrect) and high computational time for wideband, high-order systems. The grey-box process begins with grey-box impedance fitting for a single inverter. This is achieved using the Vector Fitting (VF) algorithm based on discrete frequency response data. Such data is acquired via disturbance injection (e.g., series voltage or parallel current injection) and subsequent voltage and current component extraction using FFT. VF then converts this discrete data into a continuous transfer function. For multi-inverter systems, impedances are aggregated by combining "source" and "load" subsystems via series and parallel connections. The core of the proposed method lies in stability analysis based on Bode diagrams, which explicitly determines two crucial quantities that the IF method does not adequately

handle: 1. The number of RHP poles: Determined by examining Bode diagrams of load impedance (ZL) and source admittance (YS). If the phase angle corresponding to an amplitude peak exceeds the range $[-90^\circ, 90^\circ]$, the pole is in the RHP. 2. The number of Nyquist circles: For the function L_m (product of ZL and YS), this is determined by analyzing key frequency points, statistical regions, and "crossings" (positive/negative) in Bode diagrams, which indicate how the Nyquist curve encircles the $(-1, j0)$ point. The final stability criterion is given by $P(I) = P(ZL) + P(YS) + N(-1, j0)(L_m) = 0$. The method offers significant advantages over IF, being more accurate, intuitive, computationally faster, and having fewer technical barriers for engineering application.

Helman, S., Jung, H., Baeckeland, N., Ramasubramanian, D., and Dhole, S. Grey-box System Identification of Grid-forming Inverters. 2024 9th IEEE Workshop on the Electronic Grid, eGRID 2024 (2024)

This article demonstrates the effectiveness of grey-box system identification for simplifying and understanding the nonlinear dynamics of grid-forming inverter (GFM) models. Grey-box here means using system outputs, with a priori knowledge of internal structure and parameters, to "fit" (curve-fitting) nonlinear dynamics to a linear equivalent. The method relies on a reduced-order LTI model, inspired by synchronous generator swing equations, aiming to capture the frequency and power dynamics of generating resources. This model includes parameters such as inertia (M), damping (D), and droop (R). The identification procedure uses the Matlab System Identification Toolbox (SIT). It involves: 1. Simulation of high-order nonlinear models: GFM models (with droop, dVOC, or VSM control) and synchronous generator models are simulated, recording their frequency and power outputs. 2. Construction of idgrey object: A Matlab object specifying the simplified LTI state-space matrix structure and learnable parameters (M, D, R) is created. 3. Execution of greyest function: This SIT function fits the idgrey LTI model to simulated outputs, using a "regression fit" based on a weighted sum-of-squares error. 4. Comparison: The learned LTI system outputs are compared with the original nonlinear system (using normalized RMS comparison for fit quality). 5. Parameter Correspondence: Learned LTI parameter values are compared to corresponding parameters of the high-order nonlinear models. A key finding is that for individual GFMs, inertia and droop parameters in high-order models proved to be reliable predictors of M and R parameters in the LTI model. However, when applied to a network of connected GFMs, while overall system performance in frequency

and generation response could be accurately matched by the LTI model, the underlying resource parameters were ** significantly distorted** in the learned model. This suggests the need for further work on LTI models that can explicitly capture network topology for a more accurate fit of transient dynamics.